

PRAJAL JAIN

AI & Machine Learning · Computer Vision on Embedded Hardware

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SUMMARY

I came to AI from mechatronics, which means I have spent as much time chasing a wiring fault as a training loss. I build computer vision that runs on Raspberry Pi and Arduino, inside real power and timing limits — and when a system fails in the field I can usually tell whether it is the model, the lighting or the mounting. My work has been demonstrated to Indian Army Southern Command and run live in front of crowds of 5,000+.

TECHNICAL SKILLS

AI & Machine Learning	Python, TensorFlow, TFLite, CNNs, Deep Learning, Reinforcement Learning, Hyperparameter Tuning, Model Evaluation
Computer Vision	OpenCV, MediaPipe, YOLOv8, Object Detection, Image Augmentation, Image Processing, On-Device Inference
Robotics & Embedded	Raspberry Pi, Arduino, Embedded C / C++, MPU-6050 IMU, PID Control, Servo & ESC Control, I ² C, PWM
Data & Tooling	Pandas, NumPy, SQL, Power BI, Git, Linux / Shell Scripting, Streamlit

PROJECTS

AI Crowd Detection & Headcount Monitoring

[AI-Crowd-Monitor](#) · YOLOv8, OpenCV

- YOLOv8 person detection with live headcount and automated email alerts on threshold breach.
- Deployed at Shri Mahalaxmi Temple, Pune during Navratri 2025, peak crowds above 5,000.**
- Split detection into count-then-draw passes, fixing a bug where box colours were assigned by detection order rather than crowd size.

AI-Powered Payload Drone

[CNN](#), [Raspberry Pi](#), [OpenCV](#)

- Built an 8-rotor heavy-lift octocopter, 100 kg payload class, with 6–8 minute flight endurance.
- Implemented CNN obstacle detection on Raspberry Pi as a dedicated collision-avoidance layer, independent of the flight controller handling stabilisation.
- Demonstrated to Indian Army Southern Command.** Institutional project; code restricted.

AI Waste Sorting System

[ai-waste-sorting](#) · [MediaPipe](#), [TFLite](#), [Pi](#)

- Custom TFLite detector classifying plastic, paper and metal at 92–96% accuracy on-device, sorting one item every 4–5 seconds.
- Designed the mechanism — funnel, holding tray, rotating drum with radial segments — driven by two servos on separate PWM channels.
- Rebuilt the vision loop as producer/consumer threads, removing a 7-second freeze per item caused by servo movement inside the detection callback.

Arduino Quadcopter — Tuned Flight Controller

[arduino-quadcopter-tuned](#) · [Embedded C](#), [PID](#)

- 450-class build, 1.5 kg all-up, 1400 KV motors, 3S 3100 mAh, ~15 min endurance, MPU-6050 on a 250 Hz loop.
- Added a receiver failsafe the stock firmware lacks** — an RC receiver holds its last output on signal loss, so stick values cannot detect it; the ISR timestamps every pulse edge and 500 ms of staleness is treated as link loss.
- Fixed a latent bug where the gyro integration constant was hardcoded to the ± 500 dps range, silently corrupting angle calculation whenever full-scale changed.
- Fork of YMFC-AL v1.4 by Joop Brokking.

PROFESSIONAL EXPERIENCE

AI & Industrial Data Analyst Intern

Jun 2026 – Present

Rishabh Instruments Limited

Nashik, India

- Developed predictive models on machine telemetry to flag calibration drift before it caused downtime, working with continuous sensor streams from production equipment.
- Engineered the feature pipeline for that telemetry — cleaning, resampling and outlier handling on voltage, current, temperature and humidity channels.
- Built 11 interactive Power BI dashboards for real-time shop-floor monitoring, replacing roughly 7.5 hours of manual reporting per week.

Machine Learning Engineer Intern

Aug 2025 – Dec 2025

Naviksha Technologies

Remote

- Trained a reinforcement learning agent on sensor data to complete circuit laps autonomously, reaching an 89% lap-completion rate over evaluation episodes.
- Built a supervised image classification model on X-ray and MRI scans to detect bone fractures and ligament tears, 85% validation accuracy (research, not clinical use).
- Ran 13 experiments across reward functions and hyperparameters, and built the pre-processing pipeline — normalisation, outlier handling, image augmentation.

EDUCATION

B.Tech in Artificial Intelligence

2024 – 2027

Vishwakarma University, Pune

GPA: 8.65 / 10

Diploma in Mechatronics Engineering

2021 – 2024

Indo-German Tool Room (IGTR), Chh. Sambhajinagar

Distinction

CERTIFICATIONS

DeepLearning.AI: Neural Networks and Deep Learning, Improving Deep Neural Networks, Structuring ML Projects (2025) **Google:** Data Analytics Professional Certificate (2025) **IBM:** Databases and SQL, with Honors (2025)